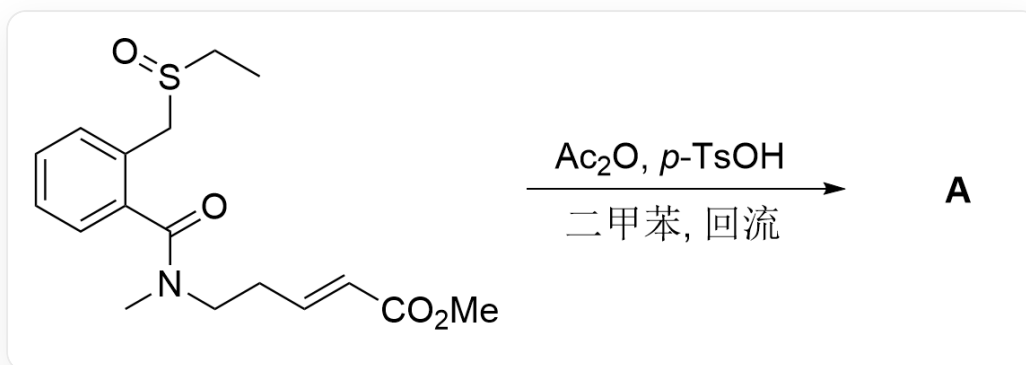


Question

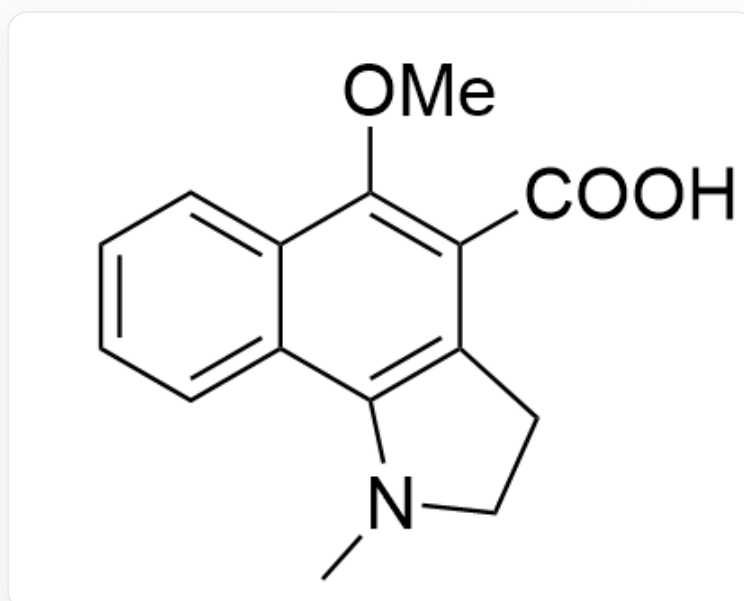


CCS(CC1=CC=CC=C1C(N(CC/C=C/C(OC)=O)C)=O)=O>CC1=CC=C(S(=O)(O)=O)C=C1.CC(OC(C)=O)=O,xylene,reflux>[A],**A** is the product

Given that the molecular formula of the reaction product **A** is $\text{C}_{15}\text{H}_{15}\text{NO}_3$, and **A** contains three rings, provide the structural formula of **A**.

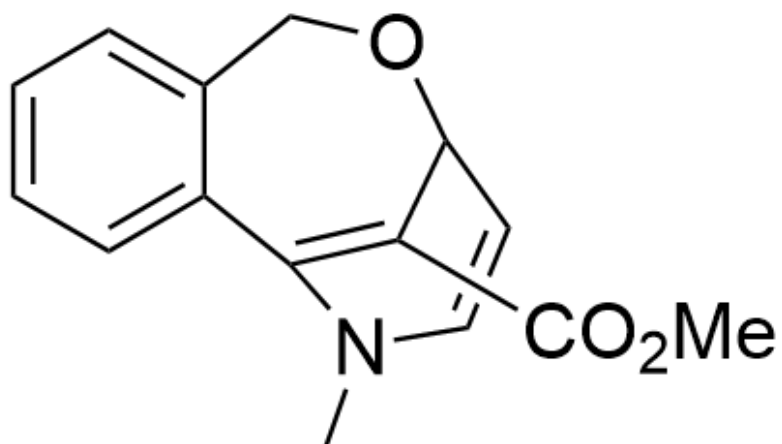
A. All other options are incorrect

B.



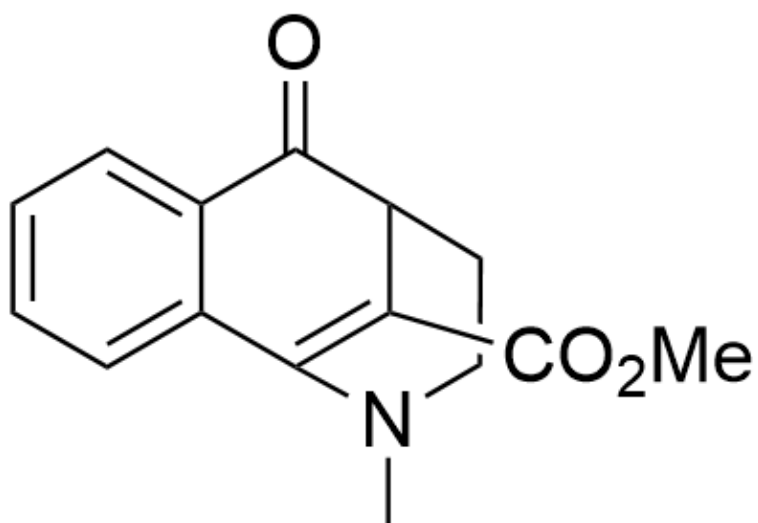
CN(CC1)C2=C1C(C(O)=O)=C(OC)C3=CC=CC=C32

C.



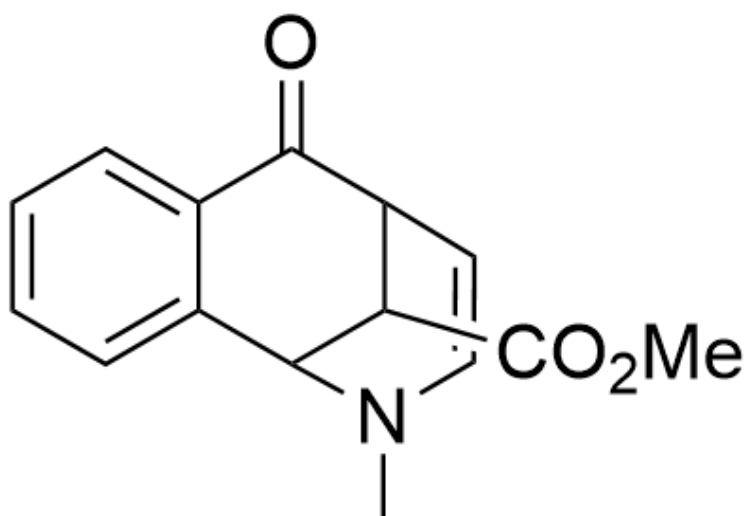
CN1C(C2=CC=CC=C2CO3)=C(C(OC)=O)C3C=C1

D.



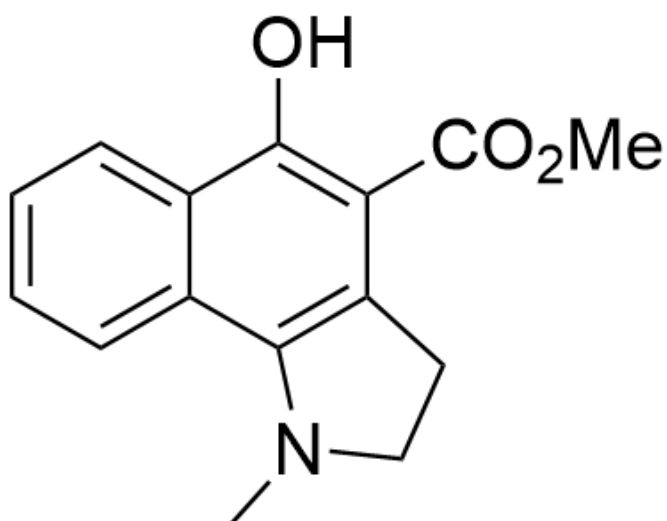
CN1C(C2=CC=CC=C2C3=O)=C(C(OC)=O)C3CC1

E.



CN(C=CC1C2C(OC)=O)C2C3=CC=CC=C3C1=O

F.



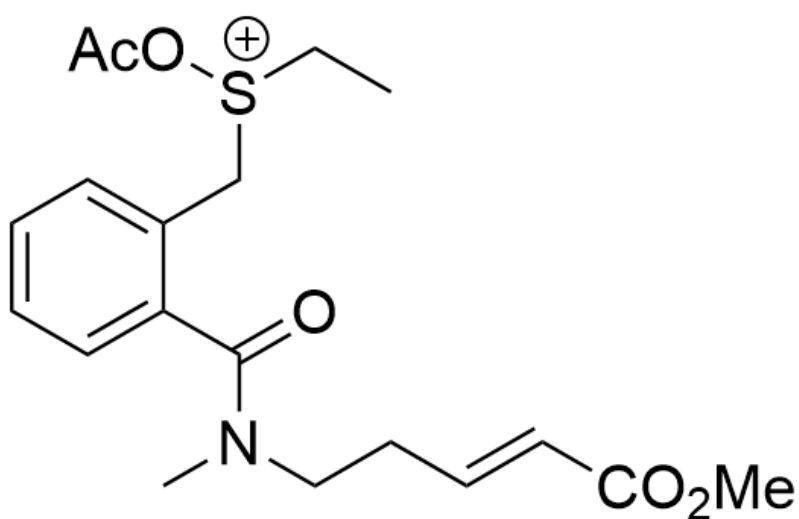
OC1=C(C(OC)=O)C2=C(N(C)CC2)C3=CC=CC=C31

Answer

Correct Answer: **F**

Detailed Explanation

First, the oxygen atom in the sulfoxide is acetylated to obtain intermediate **1**



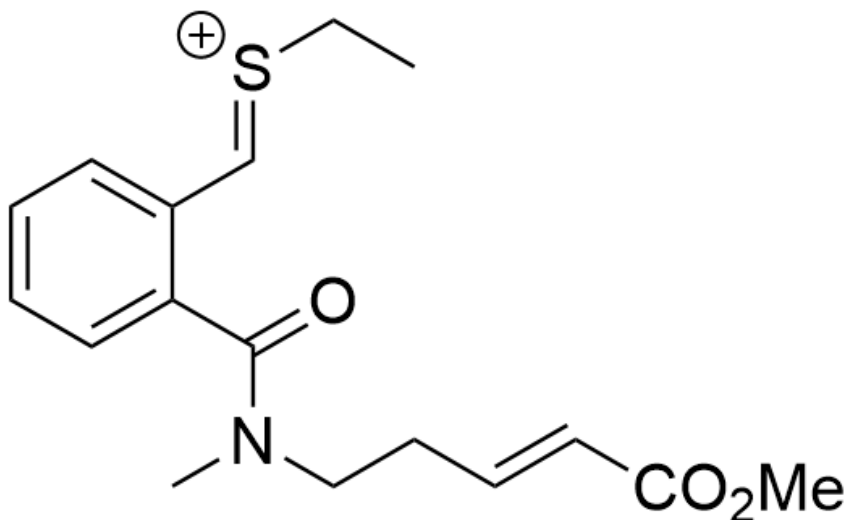
Intermediate **1**: CC[S+](OC(C)=O)CC1=CC=CC=C1C(N(CC/C=C/C(OC)=O)C)=O

CHECKPOINT

1 PTS

Intermediate **1**: CC[S+](OC(C)=O)CC1=CC=CC=C1C(N(CC/C=C/C(OC)=O)C)=O

Next, elimination of one molecule of acetic acid yields the reaction intermediate **2**



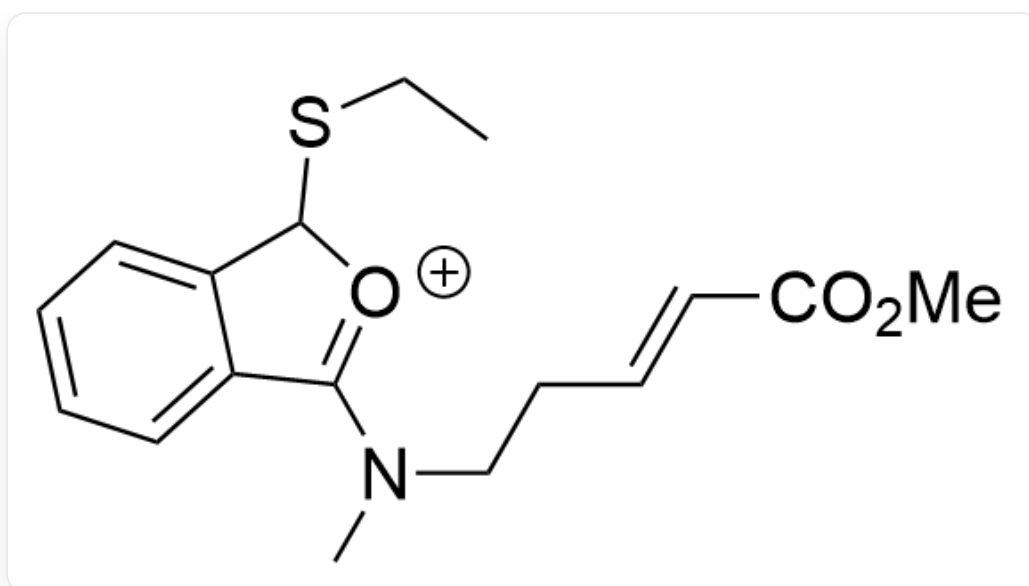
Intermediate **2**: CC/[S+]=C/C1=CC=CC=C1C(N(CC/C=C/C(OC)=O)C)=O

CHECKPOINT

1 PTS

Intermediate **2**: CC/[S+]=C/C1=CC=CC=C1C(N(CC/C=C/C(OC)=O)C)=O

The adjacent oxygen atom nucleophilically attacks the carbocation to form a five-membered ring, yielding intermediate **3**



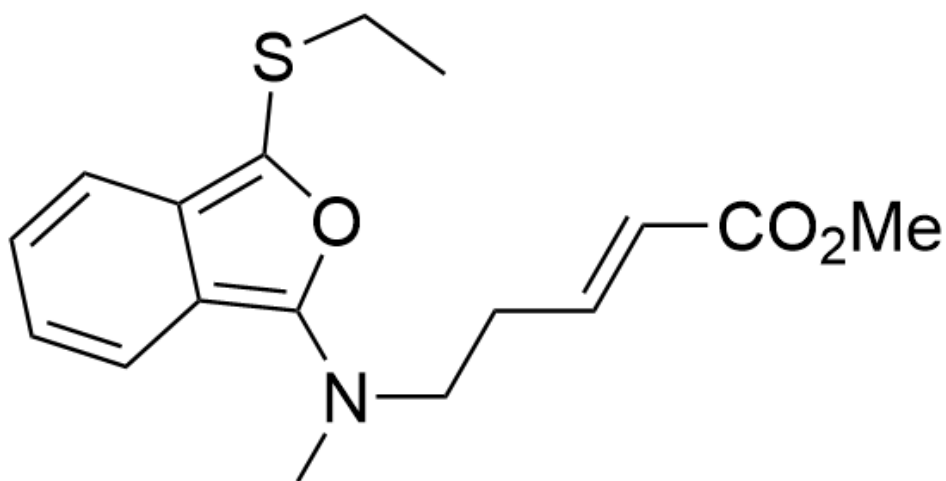
Intermediate **3**: CCSC1C2=CC=CC=C2C(N(C)CC/C=C/C(OC)=O)C=[O+]1

CHECKPOINT

1 PTS

Intermediate **3**: CCSC1C2=CC=CC=C2C(N(C)CC/C=C/C(OC)=O)C=[O+]1

Next, deprotonation and aromatization yields intermediate **4**



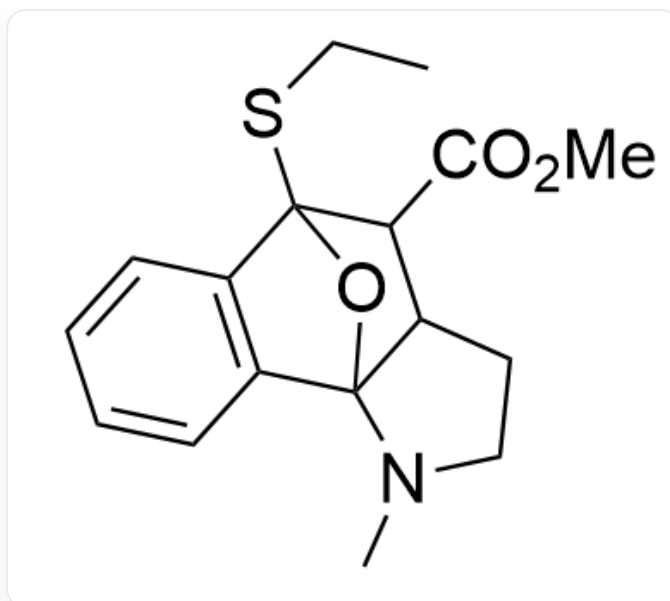
Intermediate 4: CCSC1=C2C(C=CC=C2)=C(O1)N(CC/C=C/C(OC)=O)C

CHECKPOINT

1 PTS

Intermediate 4: CCSC1=C2C(C=CC=C2)=C(O1)N(CC/C=C/C(OC)=O)C

Next, an intramolecular DA reaction occurs to give intermediate 5



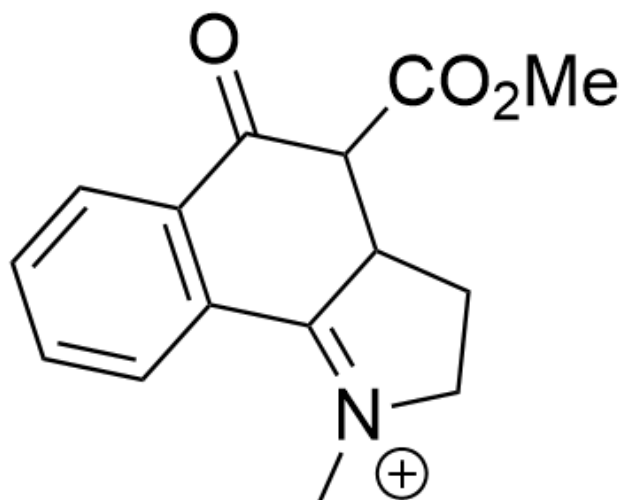
Intermediate **5**: CCSC12C3=C(C4(O1)N(CCC4C2C(OC)=O)C)C=CC=C3

CHECKPOINT

1 PTS

Intermediate **5**: CCSC12C3=C(C4(O1)N(CCC4C2C(OC)=O)C)C=CC=C3

Then, elimination of one molecule of thiolate anion yields intermediate **6**



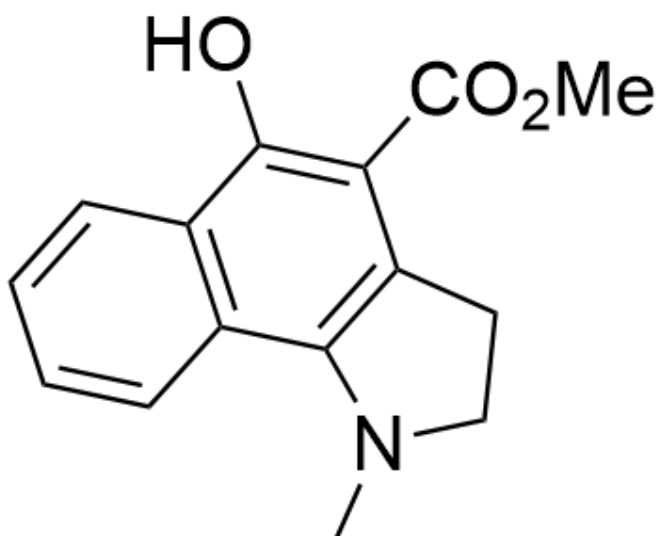
Intermediate **6**: O=C1C2=C(C3=[N+](CCC3C1C(OC)=O)C)C=CC=C2

CHECKPOINT

1 PTS

Intermediate **6**: O=C1C2=C(C3=[N+](CCC3C1C(OC)=O)C)C=CC=C2

Finally, intermediate **6** undergoes aromatization to yield product **A**.



OC1=C(C(OC)=O)C2=C(N(CC2)C)C3=C1C=CC=C3